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Vehicle level health and usage monitoring of electric unicycles using end user telemetry

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Abstract

Electric unicycles (EUCs) have evolved into high-performance personal vehicles capable of daily commuting at substantial speeds and energies, yet they remain largely outside the scope of established vehicle health and usage monitoring practices. We interpret vehicle-level health as the ability of the combined battery and power stage to sustain performance with acceptable voltage and thermal margins under representative operating conditions — a practical definition tailored to the limited telemetry available on commercially deployed EUCs. Sudden loss of propulsion, often suspected to be linked to electrical or thermal limitations, can lead to severe rider injuries, while existing mobile applications focus mainly on real-time dashboards and trip logging without providing long-term, vehicle-level health assessment. This paper introduces EUC_SOH, a vehicle health and usage monitoring system for electric unicycles that operates on ride logs collected over several thousands of kilometers. EUC_SOH derives empirical, physically interpretable indicators from pack-level telemetry and usage data, including a telemetry-derived equivalent resistance proxy, battery-dominated resistance normalized to 25°C, power MOSFET resistance at operating temperature, voltage sag metrics, and current and temperature-based stress indicators. An optional stylized ageing simulator and dedicated anomaly and degradation detection modules, including CUSUM-like change detection, are used to characterize both slow degradation trends and rare out-of-family events at the vehicle level under realistic data constraints. We evaluate EUC_SOH on real-world datasets from multiple EUCs, complemented by simulated ageing and fault scenarios. Results from the real-world logs show stable or changing intra-wheel indicator trajectories over several thousand kilometres, while the controlled synthetic scenarios demonstrate detection of major injected deviations over tens of thousands of simulated kilometres. We discuss how such vehicle-level health indicators, built from end-user-accessible logs, could support safer operation, early maintenance decisions and more systematic consideration of electric unicycles as serious vehicles within the broader framework of health and usage monitoring systems.

Keywords Electric unicycles, Micromobility safety, Health and usage monitoring systems, Telemetry-based health indicators, Power electronics degradation, Condition-based maintenance



1 Introduction

Personal electric vehicles such as electric unicycles (EUCs) have rapidly evolved from niche gadgets to high-performance transportation devices capable of sustaining daily commuting and interurban trips at substantial speeds and energies. Yet, EUCs largely remain outside the scope of traditional vehicle safety engineering and health and usage monitoring practices, even though a single loss of propulsion typically results in immediate loss of balance and a high risk of injury for the rider.

Existing national and international reports on micromobility safety mainly consider aggregated categories such as e-scooters, e-bikes and other motorised micromobility devices, and do not isolate electric unicycles as a separate class [1–3]. In France, the annual road safety reports from ONISR document a marked increase in crashes, serious injuries and fatalities among micromobility users since 2019, but without an EUC-specific breakdown of causes or failure modes [1]. In the United States, the Consumer Product Safety Commission reports similar trends for micromobility products over the 2017–2023 period, again without distinguishing EUCs within this group [2]. At European level, ETSC analyses e-scooter safety and other micromobility devices in aggregate, calling for improved data collection and mandatory safety standards, but does not provide EUC-specific statistics [3]. To the best of our knowledge, there is therefore no official dataset that systematically separates EUC incidents caused by intrinsic hardware or powertrain failures from those caused by collisions or loss-of-control. For EUCs, suspected sudden loss-of-power events are mostly reported informally in online communities, typically without access to detailed vehicle logs or structured post-mortem analysis.

Compared to other personal electric vehicles (PEVs) such as e-scooters or skateboards, electric unicycles have a much steeper learning curve: riding safely requires active balance control in both pitch and roll, and beginners must overcome a non-trivial entry barrier before reaching basic proficiency. Once this barrier is passed, however, EUCs offer a unique combination of versatility and capability: large wheels, high torque and, on recent models, long-travel suspension allow riders to negotiate curbs, potholes, rough tracks and off-road sections without dismounting, while sustaining urban cruising speeds and long commutes on a single vehicle. In practice, a single EUC can cover use cases that would otherwise require multiple PEVs (urban scooter, trekking bike, light off-road vehicle) or even a car, which explains why experienced riders tend to push these machines close to their electrical and thermal limits in real-world conditions. The main controller board acts as the vehicle's nerve center: it measures pack voltage/current (via embedded sensors), monitors temperatures, and dynamically allocates power between the battery and motor for traction or regenerative braking. These measurements, exposed via BLE, enable long-term monitoring despite the lack of direct access to cell-level data.

By contrast, in other transportation domains, Health and Usage Monitoring Systems (HUMS) and vehicle health monitoring systems are routinely used to collect on-board sensor data, derive health indicators and support condition-based maintenance for safety-critical vehicles, including military land platforms [4–7]. These systems rely on long-term usage data and diagnostic algorithms to detect degradation trends and rare anomalies, thereby reducing unexpected failures and improving availability. In parallel, a large body of work on lithium-ion batteries has shown that capacity fade and the increase of DC internal resistance are major contributors to state-of-health (SoH) reduction, and

that resistance or sag under load provide robust health indicators when compared under comparable operating conditions [8–11]. In this work, we interpret state of health (SoH) at the vehicle level as the ability of the combined battery pack and power stage to store and deliver electrical energy with acceptable voltage and thermal margins, compared to the same system when new, as reflected by the long-term evolution of equivalent resistance, sag and minimum under-load voltage. Recent studies have further emphasized the importance of non-linear ageing trajectories with distinct elbows or regimes of accelerated degradation in internal resistance and related metrics, and their implications for failure risk and end-of-life decisions [12, 13]. For power MOSFETs and other power semiconductor devices, changes in on-state resistance, junction temperature and related electrical parameters have been widely investigated as precursors of degradation and failure in accelerated ageing, power-cycling and condition-monitoring studies [14–19]. However, to the best of our knowledge, no HUMS-like framework has been proposed and evaluated for electric unicycles considered as everyday vehicles, nor has the combined long-term evolution of battery and power electronics indicators been systematically studied at the vehicle level for this class of devices.

Rather than aiming to reproduce laboratory-grade SoH estimation; which would require controlled pulse protocols, per-cell measurements and calibrated ground truth; EUC_SOH is designed as a pragmatic health monitoring methodology for vehicles where only pack-level, end-user-accessible telemetry is available. This positions it in the growing body of work on data-driven, field-deployable health indicators for electrified vehicles, rather than in the tradition of bench-based SoH benchmarking.

While EUC failures can stem from mechanical issues (e.g., bearing wear, structural fatigue) or electronic faults (e.g., BMS malfunctions, sensor drift), this work focuses on battery and power MOSFET degradation for three reasons:

1. Criticality: Sudden loss of propulsion in EUCs is frequently suspected to be associated with electrical/thermal limits in the battery or power stage (as reported in user forums and incident analyses).
2. Data availability: Pack-level voltage, current, and temperature telemetry are consistently exposed by EUC companion apps, enabling long-term monitoring.
3. Literature gap: Prior PHM work on micromobility has not addressed these components at the vehicle level.

In this work, we introduce *EUC_SOH*, a vehicle-level health and usage monitoring system for electric unicycles that offers users insights about the internal status of their devices. *EUC_SOH* operates on ride logs collected over several thousands of kilometers, using signals available to end-users (pack-level telemetry, motor and controller temperatures, usage patterns) to monitor the health of the energy and power stages, including both the battery pack and the power MOSFET stage. The system combines interpretable feature extraction, an ageing simulator and anomaly/degradation detection mechanisms to identify subtle deviations and rare major events in long-term field data, at both slow-degradation and single-outlier scales. We show on real-world EUC datasets and simulated ageing scenarios that *EUC_SOH* can provide potentially useful health indicators for maintenance screening and risk awareness for EUCs.

Unlike traditional PHM systems that rely on fixed baselines (e.g., fleet-wide nominal values), *EUC_SOH* uses vehicle-specific, adaptive baselines derived from each wheel's

historical data. This choice is driven by the high variability in EUC usage conditions (rider mass, terrain, ambient temperature), which would make global thresholds either overly conservative or ineffective for individual vehicles.

Although this work focuses on electric unicycles as a particularly demanding and under-studied class of personal electric vehicles, the overall approach is not EUC-specific. Any micromobility platform that exposes comparable pack-level telemetry (voltage, current, temperatures) and trip logs through its companion application or telematics backend could, in principle, host a similar vehicle-level health and usage monitoring layer. The main constraints are the availability and quality of the exposed signals, not the details of the vehicle architecture, which suggests that the concepts explored here may be transferable to other PEVs such as e-scooters and e-bikes as their logging capabilities mature.

The proposed method is evaluated through: (i) empirical comparison of resistance estimation techniques, (ii) robustness analysis of statistical thresholds to outliers, and (iii) controlled experiments on synthetic ageing scenarios demonstrating early fault detection.

This work makes three contributions.

First, it introduces EUCSOH, a pragmatic vehicle-level health and usage monitoring methodology for electric unicycles that operates solely on end-user-accessible pack-level telemetry and ride logs, without assuming laboratory-grade instrumentation or controlled test protocols. Second, it provides a long-term field dataset and analysis pipeline for multiple wheels, including interpretable indicators such as Equivalent Load Proxy (ELP), battery-attributed resistance, MOSFET-attributed resistance, sag metrics and stress integrals, combined with vehicle-specific baselines and simple change detectors. Third, it presents exploratory case studies and stylized ageing scenarios illustrating how these indicators reveal non-trivial degradation patterns and rare out-of-family events under realistic usage constraints, while explicitly acknowledging the lack of independent laboratory ground truth and the limited sample size of this initial study.

2 Background and related work

2.1 Electric unicycles in the literature

Existing academic work on electric unicycles has largely focused on structural design and control for self-balancing and trajectory tracking. Early and more recent contributions treat the EUC as a challenging case for nonlinear control and mobile robotics, addressing modelling, stabilisation and path-following controllers, often validated on laboratory prototypes [20–23]. These studies demonstrate that EUCs are technically viable and controllable, but they do not consider long-term field usage, health monitoring of the energy and power stages, or safety implications of component degradation.

A second line of work has explored prototype design and feasibility of electric unicycles as alternative urban mobility options, sometimes emphasising potential benefits in terms of congestion and local pollution [20]. However, these prototypes are not evaluated as long-lived vehicles with monitored health, and do not address battery or power electronics degradation under real usage over thousands of kilometers. To the best of our knowledge, no prior work has treated electric unicycles as everyday vehicles equipped with health and usage monitoring comparable to HUMS in other transport

domains, nor has any study exploited real-world EUC logs over thousands of kilometers to jointly monitor battery and power electronics health.

This scoped literature review was updated through August 2026 using the search terms “electric unicycle”, “health monitoring”, “health and usage monitoring”, “state of health”, “battery degradation”, “power electronics”, “micromobility”, and “prognostics” in Google Scholar.

2.2 Micromobility, climate and safety

Micromobility devices such as e-scooters, e-bikes and other PEVs are increasingly promoted as low-carbon alternatives to private cars for short and medium-distance trips, with potential benefits in terms of greenhouse gas emissions, local air pollution and congestion when they effectively replace motorized trips [2, 3]. Life-cycle assessments of shared and private micromobility services show that their environmental performance is highly sensitive to vehicle lifetime, utilisation and operational practices; longer-lived devices with high mileage and low empty rebalancing tend to offer the largest climate benefits [11, 24]. Electric unicycles, with their low vehicle mass relative to the rider and high energy efficiency per kilometre, have the potential to fit particularly well into this picture, although they are rarely considered explicitly in micromobility environmental assessments.

From a safety perspective, national and international reports document a marked increase in crashes, serious injuries and fatalities involving micromobility users since 2019 without a proper breakdown by PEV category [1–3] as discussed in Sect. 1. Most policy and research efforts to date have focused on infrastructure, speed regulation, user behaviour and protective equipment, while the internal health of the vehicles themselves has received comparatively little attention. In particular, there is no systematic assessment of how long-term degradation of batteries and power electronics might affect the risk of loss-of-power events for micromobility vehicles, despite the fact that these devices are increasingly sensorised and connected.

Current regulatory approaches to micromobility have largely emphasized speed limits, access rules and equipment requirements, often with limited empirical insight into how these measures interact with the technical reliability and degradation of the vehicles themselves. A more integrated perspective that combines infrastructure, user behaviour and vehicle health could help move beyond purely heuristic rules.

2.3 Health and usage monitoring systems for vehicles

In contrast, Health and Usage Monitoring Systems (HUMS) and related vehicle health monitoring systems have been widely studied and deployed in other transport domains. In aerospace and military land vehicles, HUMS architectures combine on-board sensing, data logging and diagnostic algorithms to monitor critical subsystems, support condition-based maintenance and reduce unexpected failures [4–7]. Recent work on vehicular sensor networks and data analytics further emphasizes the role of long-term usage data and prognostic indicators in managing fleets of safety-critical vehicles [4].

Despite conceptual similarities between micromobility vehicles and other electric vehicles in terms of energy storage and powertrain architecture, we are not aware of any HUMS-like framework specifically designed for electric unicycles or other EDPMs. Existing micromobility deployments typically log trips for billing and basic statistics, but

do not implement systematic health indicators for the battery pack or power electronics, nor do they leverage long-term field data to characterize degradation at the vehicle level. This gap motivates the development of vehicle-level health and usage monitoring tools tailored to the constraints and data channels of commercially available EUCs.

2.4 Battery and power electronics health monitoring from field data

For lithium-ion batteries, a substantial body of work has investigated state-of-health estimation and fault detection using field data from electric vehicles, stationary storage systems and dedicated test benches. Internal resistance, voltage sag under load and derived features have been repeatedly identified as informative health indicators, especially when analyzed under comparable operating conditions [8–11]. Recent studies have highlighted the presence of elbows or knees in ageing trajectories, corresponding to regimes of accelerated degradation and rapid loss of performance, and have explored data-driven methods for detecting abnormal SoH estimates and battery faults in large fleets of electric vehicles [12, 13, 25–29].

In parallel, power MOSFETs and other power devices have been the subject of extensive reliability and condition monitoring research. Variations of on-state resistance, junction temperature and related electrical parameters have been used as precursors of degradation under thermal cycling and accelerated ageing, and have enabled remaining useful life estimation in controlled experiments [14–19]. However, most of these studies target automotive inverters, power modules or individual battery packs, and are not integrated into a vehicle-level health and usage monitoring framework for micromobility devices. To the best of our knowledge, no prior work has exploited long-term EUC ride logs over thousands of kilometers to jointly monitor the health of the battery pack and the power MOSFET stage, and to relate their evolution to vehicle-level safety margins.

3 System description and methods

The datasets and source code used to generate the results of this paper are openly available at the EUC_SOH Zenodo repository <https://doi.org/10.5281/zenodo.21861254>, with EUC brands/models and personally identifying information removed from the logs.

Figure 1 provides a high-level overview of the EUC SOH pipeline, from raw ride logs to per-ride health indicators, per-wheel statistical baselines, change detection and alarm reporting.

3.1 Telemetry and log structure

EUC_SOH operates on ride logs exported from user-facing applications such as EUC World or WheelLog. Each log corresponds to a single ride and contains timestamped

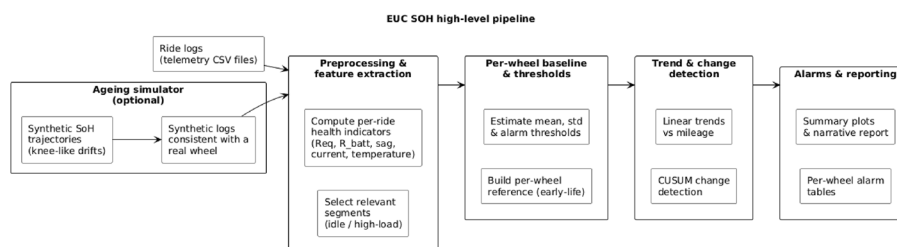


Fig. 1 High-level overview of the EUC_SOH pipeline, from ride logs and synthetic ageing scenarios to per-ride health indicators, per-wheel baselines, trend/change detection results, and alarm/report outputs

samples of pack voltage, battery current, vehicle speed and, when available, board and motor temperatures and phase current. For each wheel, we consider a collection of CSV logs covering several hundreds to thousands of kilometers, and we use the wheel odometer value recorded in each file to position every ride on the lifetime axis.

EUC companion apps (e.g., EUC World, WheelLog) interface with the vehicle's main controller board, which continuously measures battery (via shunt or Hall-effect sensors) and temperature (via onboard thermistors). The controller also manages power flow between the battery pack and the motor, handling both traction (discharging the pack to propel the wheel) and regenerative braking (charging the pack during deceleration). These measurements are broadcast over Bluetooth Low Energy (BLE) at 1–10 Hz and logged by the app, forming the raw telemetry used by EUC_SOH.

Within each log, we focus on periods of high load where the vehicle is simultaneously moving and drawing significant current. Concretely, we retain samples that satisfy a speed threshold (typically 20 km/h) and an adaptive threshold on the absolute battery current, and discard low-load segments (near-idle coasting or very low current). This selection aims to approximate quasi-steady high-load conditions where voltage sag and Joule losses are most informative about the combined health of the battery and power electronics.

3.2 Equivalent load proxy and global health indicator

For each log, EUC_SOH first identifies quasi-idle segments where the vehicle is moving slowly or not at all and the battery current remains close to zero. On these segments, it estimates a local idle pack voltage V_{idle} as a high percentile (typically 95th) of the measured voltage at very low current. When a reliable voltage-based state-of-charge (SoC) proxy is available for the wheel (inferred from the maximum pack voltage, number of series cells and near-full resting voltage), these quasi-idle plateaus are also associated with an approximate SoC value, and a piecewise-linear profile $V_{\text{idle}}(\text{SoC})$ is constructed from them. For any sample in the log, a local idle voltage $V_{\text{idle,local}}$ is then obtained either by evaluating this profile at the corresponding SoC or, when such a profile cannot be built, by falling back to an average of the quasi-idle voltages.

Figure 2 illustrates the relationship between the telemetry-derived local idle voltage and the voltage-based SoC proxy for one representative wheel (Wheel Z). The quasi-idle plateaus extracted from real ride logs align along a smooth, monotonic open-circuit-voltage-like curve, while the binned median profile remains stable despite the natural dispersion of field data. This supports the interpretation of $V_{\text{idle,local}}$ as a practical OCV proxy under end-user telemetry constraints, and justifies its use as the reference voltage in the computation of sag and ELP.

Using this local reference, EUC_SOH focuses on periods of high load where the vehicle is simultaneously moving and drawing significant current. Concretely, we retain samples that satisfy a speed threshold (typically 20 km/h) and a current window adapted to the pack voltage (with a minimum threshold; if too few points are found, the window is widened), and we optionally restrict the analysis to mid-range SoC (e.g. between 20% and 90%) when SoC information is available. Under high-load, quasi-steady conditions (speed > 20 km/h, stable current). We refer to this quantity as the Equivalent Load Proxy (ELP) rather than as a resistance, to emphasize that it is a derived, system-level indicator

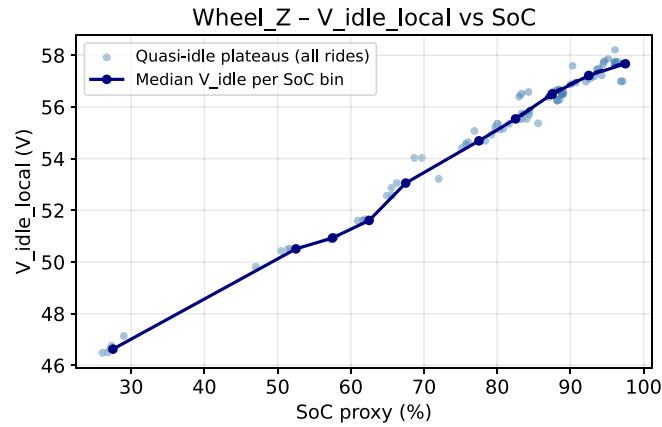


Fig. 2 Relationship between the telemetry-derived local idle voltage $V_{\text{idle,local}}$ and the voltage-based SoC proxy for Wheel Z. Each point corresponds to a quasi-idle plateau extracted from real ride logs, defined by low current and stable voltage over several seconds. The solid line shows the median $V_{\text{idle,local}}$ in successive SoC bins. The resulting smooth, monotonic profile supports the use of $V_{\text{idle,local}}$ as a practical open-circuit-voltage-like reference for the computation of sag and Equivalent Load Proxy (ELP) under field conditions

that bundles the ohmic contributions of cells, busbars, connectors and power devices, and is not equivalent to a laboratory-grade DC internal resistance measurement:

$$\text{sag} = V_{\text{idle,local}} - V, \quad \text{ELP} = \frac{\text{sag}}{|I|}, \quad (1)$$

where $V_{\text{idle,local}}$ is the open-circuit voltage estimate. This approximation is valid for slow dynamics (sampling rate: 1–10 Hz) and aligns with DC resistance estimation methods in battery PHM [8, 9].

We then aggregate these pointwise values at the log level using the median and the 95th percentile:

- ELP_median : median of ELP over high-load samples;
- ELP_95p : 95th percentile of ELP. ELP_median is used as the main global health indicator for the energy and power stages combined (battery, MOSFETs and wiring). Numerous studies have shown that DC internal resistance and voltage sag increase with ageing and closely track capacity fade and performance loss in lithium-ion batteries, especially in the early and mid-life regions [8–11, 24]. By construction, ELP_median captures the dominant ohmic contribution of the pack under representative load conditions for each ride, while the use of a local $V_{\text{idle,local}}$ profile reduces the bias that would arise from large state-of-charge swings within a single log.

It is important to emphasize that ELP and ELP_median are *pack-level indicators* rather than direct estimates of a specific electrochemical parameter. They lump together the ohmic contributions of battery cells, busbars, connectors and power devices, and they are inferred under dynamic load and finite sampling frequency, not under controlled laboratory DC conditions. As such, they should be interpreted primarily as relative indicators within the history of a given wheel, under comparable operating conditions, rather than as precise absolute measurements of cell-level internal resistance. This empirical nature is unavoidable given the limited telemetry exposed by commercially available

EUCs, but it remains sufficient for detecting large intra-wheel drifts and out-of-family behaviour.

3.3 Validation of the static resistance approximation

In this work, we use the term high-load event (or informally pulse) to refer to a contiguous segment of samples satisfying simultaneously: speed > 20 km/h, battery current above the adaptive threshold, and an approximate SoC within the mid-range window when available. These events typically last 1–30 s in practice, corresponding to sustained acceleration or uphill segments in real EUC rides. No artificially designed current pulses are applied: the selection relies entirely on naturally occurring high-load phases in field data.

The Equivalent Load Proxy ELP in Eq. 1 is computed as a static approximation $ELP = \frac{V_{idle} - V}{|I|}$ under quasi-steady high-load conditions. This approach contrasts with classical battery resistance estimation methods, which typically rely on pulse-based techniques such as:

- Direct Current Internal Resistance (DCIR): Applied via short high-current pulses (e.g., 1–2C for 1–10 s) in controlled environments, with resistance derived from the voltage drop $\Delta V / \Delta I$ [30].
- Hybrid Pulse Power Characterization (HPPC): Standardized in automotive applications (e.g., DOE/USCAR), using defined charge/discharge pulses to extract ohmic resistance [31].
- Differential methods: Commonly implemented in Battery Management Systems (BMS) as $R = \Delta V / \Delta I$ between consecutive samples, assuming high-frequency sampling (> 100 Hz) to minimize noise and dynamic effects [32, 33].

However, EUC telemetry presents key constraints that render these classical methods impractical:

- Low sampling rate (1–10 Hz): Insufficient to capture fast dynamics or avoid division by near-zero ΔI .
- No controlled pulses: Ride logs contain arbitrary current profiles (no designed HPPC/DCIR pulses).
- Noisy sensors: Consumer-grade EUC telemetry has typical noise of ± 0.1 A (current) and ± 0.2 V (voltage), dominating ΔV and ΔI at low sampling rates.

To empirically validate our static approximation, we compared it against a differential method ($R_{eq,diff} = \frac{V_k - V_{k-1}}{I_k - I_{k-1}}$) on all 8 wheels vehicles in our dataset (90 rides, 15,800 km). For each ride, we:

- Computed $R_{eq,static}$ for high-load samples (speed > 20 km/h, $|I| > 20$ A).
- Computed $R_{eq,diff}$ for samples where $|\Delta I| > 0.1$ A (to avoid division by near-zero).
- Evaluated both Pearson (linear) and Spearman (monotonic) correlations.

Results (Table 1):

Key observations:

- No correlation: Both Pearson ($r \approx -0.01$) and Spearman ($\rho \approx 0.04$) correlations are close to zero for all wheels.

Table 1 Comparison of static vs. differential load proxy estimation methods across all wheels

Wheel	Pearson (r)	Spearman (ρ)	Noise ratio
Wheel P (134V)	0.01	0.06	4.8
Wheel E (133V)	−0.02	0.07	11.8
Wheel B (133V)	0.00	0.07	7.9
Wheel I (225V)	−0.23	−0.04	18.1
Wheel M (82V)	0.04	−0.00	17.3
Wheel S (122V)	0.00	0.03	13.8
Wheel V (82V)	0.09	0.09	8.1
Wheel Z (55V)	−0.01	0.00	9.1
Mean	−0.01	0.04	11.4

- Extreme noise: The differential method is 11.4×noisier than the static method (standard deviation ratio).
- Unphysical values: 10–30% of ELP,diff samples were negative or $> 1\Omega$, even after filtering.

Note that EUC operation never produces sustained constant-current phases at the sampling rates considered (1–10 Hz): current varies continuously with terrain, speed and rider input. Division by near-zero current is therefore not a practical concern under the high-load filter applied here, as discussed further.

Interpretation: The differential method fails on EUC field data due to:

- Low sampling rate (1–10 Hz) leading to small ΔI and division by near-zero.
- Measurement noise dominating ΔV and ΔI .
- Non-ohmic battery dynamics (e.g., polarization) violating the Ohm’s law assumption for instantaneous changes.

By contrast, our static approximation yields plausible resistance values (e.g., 0.05–0.2 Ω for typical EUC packs) and stable trends over time (Sect. 4).

Because the exact cell model, cell DCIR at the relevant state of charge and temperature, interconnect resistance, and controller topology are not available for every anonymized wheel, this calculation is used only as a broad order-of-magnitude check [34]. It does not constitute a wheel-specific physical validation of ELP.

However, we emphasize that ELP is a vehicle-level indicator combining battery, MOSFET, and wiring resistances, and should be interpreted as a relative health metric rather than an absolute physical measurement.

The static-versus-differential comparison in this section uses the full eight-wheel dataset (90 rides, $\approx 15,800$ km) to stress-test the differential estimator under realistic EUC telemetry conditions. In the subsequent results, we focus on three representative wheels and 43 logs for detailed case studies, to avoid overloading figures while still covering diverse pack voltages, wheel architectures and usage profiles.

The differential comparator at 1–10 Hz is therefore used as a stress test rather than as a ground-truth reference: it shows that naive sample-to-sample resistance estimates are dominated by measurement noise and battery dynamics under EUC telemetry constraints. We do not claim that this comparison physically validates ELP; rather, it motivates the use of a static approximation as a pragmatic, vehicle-level indicator in this data regime.

In principle, stronger alternatives could be considered, such as regression of voltage on current within short operating windows, event-based estimators, or equivalent-circuit

models fitted to each ride. However, these approaches typically require higher sampling rates, designed current pulses, or richer instrumentation than is available in commercial EUC telemetry. Exploring such models under EUC constraints is beyond the scope of this first field study and is left for future work.

3.4 Separating battery and MOSFET contributions

When additional information about the power stage is available (e.g. datasheet or measured values), EUC_SOH can separate the contributions of the battery and the MOSFET bridge to the measured Equivalent Load Proxy. The user provides:

- A total on-state resistance $R_{ds,on,25^\circ C}$ for the full MOSFET bridge at $25^\circ C$;
- A relative thermal coefficient α describing the variation of $R_{ds,on}$ with junction or board temperature;
- An optional wiring/busbar resistance R_{wiring} .

Using the maximum board/controller temperature

`temp_board_max` recorded in each log as a proxy for the MOSFET junction temperature, we estimate the hot MOSFET resistance as

$$R_{MOSFET,hot} = R_{ds,on,25^\circ C} \cdot (1 + \alpha \cdot (T_{board} - 25)) + R_{wiring}. \quad (2)$$

We then obtain a battery-only median resistance estimate by subtracting this contribution from the global median:

$$R_{batt,median} = \max(0, ELP_median - R_{MOSFET,hot}), \quad (3)$$

and we normalize battery resistance to $25^\circ C$ using an Arrhenius correction to obtain `R_batt_median_25C`.

Temperature proxy caveat. All temperature-dependent corrections in EUC_SOH rely on `temp_board_max`, the maximum temperature of the main board/controller recorded during a ride, as the sole thermal observable. This quantity is a coarse proxy: it reflects the thermal state of the controller PCB, not the junction temperature of individual MOSFETs, the surface temperature of battery cells, or the ambient temperature. The board sensor is also read at the logging frequency of the companion app (typically 1–10 Hz), so transient thermal peaks at the device level are not captured. During short high-current transients, the controller-board temperature may also lag the junction temperature of individual MOSFETs because of thermal capacitance and heat-transfer delays. The available telemetry therefore cannot resolve fast junction-temperature excursions or establish a calibrated relationship between T_{board} and T_j . As a consequence, $R_{MOSFET,hot}$ and the Arrhenius-corrected `R_batt_median_25C` should be interpreted as empirical, intra-wheel de-seasoning estimates rather than as physically exact thermal corrections. They remain useful for tracking relative changes within the history of a given wheel, but are not suitable for precise cross-wheel or cross-platform comparisons without additional thermal calibration.

The uncertainty associated with $R_{ds,on}$, its temperature coefficient, and the use of board temperature as a proxy for junction temperature directly propagates to the battery-attributed residual through Eq. 3. We therefore do not interpret the battery/MOSFET decomposition as a quantitatively validated component-level diagnosis. Its purpose in the present study is limited to providing an approximate, within-wheel attribution of

changes in the global ELP indicator. A formal sensitivity analysis over plausible electrical and thermal parameter ranges, ideally supported by controller-level measurements, is required before this decomposition can be used for quantitative cross-vehicle or component-level comparisons.

This decomposition yields three interpretable indicators per log: `ELP_median` (global), `R_mosfet_hot` (power stage), and `R_batt_median` / `R_batt_median_25C` (battery-dominated). The separation is approximate and relies on simplified thermal and electrical models, but it is consistent with prior work using on-state resistance drift as a precursor of MOSFET degradation and DC resistance growth as an indicator of battery ageing [8–12, 14–19]. It is important to note that this decomposition provides only a system-level separation, not a component-level diagnosis. The ELP indicator and its battery/MOSFET breakdown reflect the combined electrical and thermal behaviour of the pack as seen from the telemetry layer: they cannot distinguish, for instance, between a degraded cell group and a high-contact-resistance connector, or between MOSFET degradation and wiring corrosion. Users should therefore interpret rising battery- or MOSFET-attributed indicators as evidence of degradation within that subsystem boundary, not as a precise identification of the root-cause component.

3.5 Additional indicators: sag, current, temperature and phase current

Beyond Equivalent Load Proxy, EUC_SOH computes several complementary indicators for each log:

- `sag_95p`, `sag_max`: 95th percentile and maximum of voltage sag under load, capturing worst-case ohmic drops and their evolution over time;
- `v_min_strong`: minimum voltage reached under high-load conditions, reflecting the remaining safety margin before low-voltage cut-off, for a given SoC and temperature;
- `i_95p`, `i_max`: 95th percentile and maximum battery current, characterizing typical and extreme electrical stress on the pack;
- `temp_board_max`: maximum board/controller temperature, used as a proxy for MOSFET thermal stress;
- `temp_motor_max`: maximum motor temperature, when available.

When firmware provides a phase current signal, we also compute:

- `i_phase_max`, `i_phase_95p`: maximum and 95th percentile of phase current;
- `I_phase2_int`: the phase-current stress density, defined as

$$I_{\text{phase2_int}} = \frac{\int_{t_{\text{start}}}^{t_{\text{end}}} I_{\text{phase}}^2(t) dt}{T_{\text{ride}}}, \quad T_{\text{ride}} = t_{\text{end}} - t_{\text{start}},$$

expressed in A^2 (dimension of a mean squared current). When the ride duration T_{ride} cannot be inferred from the log timestamps, a fixed sampling interval of 0.1 s is used as a fallback and the raw integral $\text{A}^2 \cdot \text{s}$ is retained without normalisation.

The normalization by T_{ride} makes `I_phase2_int` comparable across rides of different durations: a long ride at moderate current and a short ride at high current producing the same thermal burden per unit time will yield the same value, whereas the un-normalized

integral would unfairly penalize longer rides. This quantity acts as a proxy for the mean power dissipated in the MOSFETs, shunts and motor windings per unit time, and thus for the sustained thermal stress applied to the power stage during the ride. This is in line with fatigue and power-cycling studies of power semiconductors, where accumulated I^2t and thermal cycling are key drivers of degradation [14, 15, 18].

The integral $\int I_{\text{phase}}^2 dt$ acts as a proxy for the total Joule energy dissipated in the MOSFETs, shunts and motor windings during the ride, and thus for the cumulative thermal stress applied to the power stage. This is in line with fatigue and power-cycling studies of power semiconductors, where accumulated I^2t and thermal cycling are key drivers of degradation [14, 15, 18].

3.6 Relationship between indicators and state of health

In this work, we treat state of health (SoH) at the vehicle level as the ability of the combined battery and power stage to deliver the power demanded by the rider without excessive voltage collapse or heat build-up under representative operating conditions. In practice, this ability is primarily governed by the effective internal resistance of the pack and by the remaining voltage and thermal safety margins.

The median Equivalent Load Proxy ELP_{median} is therefore used as the main health indicator: for a given current demand, a higher ELP_{median} means a larger voltage drop across the pack, less headroom before the firmware cut-off threshold, and higher Joule losses in the cells, wiring and power devices. A wheel whose ELP_{median} has doubled compared to its early-life baseline can still reach the same open-circuit voltage at rest, but behaves like an “old” battery that maintains nominal voltage while being unable to sustain the same power under load.

When MOSFET parameters are available, the battery-dominated resistance $R_{\text{batt,median},25^\circ\text{C}}$ refines this picture by isolating the cell contribution and normalising it to 25°C . A slow, smooth increase of $R_{\text{batt,median},25^\circ\text{C}}$ over hundreds of rides is consistent with normal cell ageing, whereas a sudden jump suggests a localised fault (e.g. damaged cell group, connector or BMS issue) rather than gradual wear. Conversely, a flat battery resistance combined with rising ELP_{median} and $\text{temp}_{\text{board,max}}$ points to power-stage or wiring degradation.

Sag-based metrics ($\text{sag}_{95\%}$, sag_{max}) and the minimum strong-load voltage $v_{\text{min,strong}}$ translate the same resistance growth into usable electrical margin: increasing sag and decreasing $v_{\text{min,strong}}$ indicate that more of the available voltage window is spent during hard acceleration, especially at low state of charge or in cold weather, thereby increasing the risk of an abrupt low-voltage cut-off. Finally, $\text{temp}_{\text{board,max}}$ and the phase-current stress integral $I_{\text{phase,int}}^2$ characterize the thermal and electrical loading history of the power stage. High values of these stress indicators do not directly measure SoH, but they explain *why* a given wheel may have reached a degraded regime more quickly than another.

By construction, all these indicators are interpreted as relative quantities within the history of a given wheel, not as absolute laboratory measurements of cell-level parameters. A high resistance or sag value in isolation does not imply that the battery is unsafe per se, only that its behavior has drifted away from its own early-life baseline by an amount that is large enough to matter for voltage and thermal safety margins.

It is important to distinguish energy-capacity SoH from power-capability SoH at the vehicle level. The indicators used here (ELP, sag under load, minimum strong-load voltage) primarily reflect the ability of the combined battery and power stage to deliver the demanded power with acceptable voltage and thermal margins, rather than the remaining energy capacity per se. A wheel whose ELP_{median} has doubled compared to its early-life baseline may still reach similar open-circuit voltages at rest, but behaves like a power-limited system under load. EUCSOH therefore focuses on vehicle-level power-capability SoH under field telemetry constraints.

3.7 Statistical reference and alarm thresholds

For confidentiality and to avoid brand-specific controversy, individual EUCs are anonymized as Wheel B, Wheel E, etc., and manufacturer/model names are not disclosed.

The statistical reference is built from each wheel's own history, using either the full log set or a low-resistance fraction as the reference subset. This approach relies on the empirically supported assumption that Li-ion and MOSFET degradation produces distinct regime changes (elbows/knees) rather than imperceptibly slow continuous drift [12, 13, 15]. Under this assumption, late-life degraded logs represent a small fraction of the total dataset, and their inclusion in the reference has limited influence on the estimated mean and standard deviation as confirmed empirically in our case studies, where late-life values exceed the danger threshold by several α even when computed on the full log set.

This behaviour is illustrated in Appendix , where strong synthetic late-life deviations on ELP_{median} for Wheel_P_bad remain a small fraction of the dataset and do not materially alter the per-wheel Gaussian bands.

On this set, we compute a mean μ and standard deviation σ for each metric. We then define:

- A green band $[\mu - 1\sigma, \mu + 1\sigma]$ representing the normal range;
- An orange band between the green band and a danger threshold;
- a danger threshold at $\mu + 2\sigma$ for metrics where higher values are worse (resistances, sag, temperatures, currents), and $\mu - 2\sigma$ for metrics where lower values are worse (minimum voltage).

These thresholds are intended as empirical screening thresholds for within-wheel monitoring and maintenance awareness. They are not safety-certified limits, manufacturer-approved operating limits, or experimentally established failure thresholds. Exceeding a threshold should therefore be interpreted as a prompt for further inspection or more cautious operation, rather than as proof that the vehicle is unsafe or that failure is imminent.

Measurement uncertainty in pack voltage and current is not specified by EUC manufacturers or companion apps. In the absence of public specifications, we assume that embedded sensors exhibit relative errors in the range of 2-5% for both voltage and current, which is consistent with the accuracy reported for typical automotive-grade Hall and shunt sensors and with field observations of pack-voltage calibration offsets reported by EUC users. Under these assumptions, a first-order error propagation on $ELP = \text{sag}/|I|$ yields a relative uncertainty on individual ELP samples on the order of 3–7%, depending on the operating point.

Two mechanisms mitigate the impact of this uncertainty on the diagnostic indicators used here. First, each per-log metric such as `ELP_median`, `ELP_95p` or `sag_95p` aggregates hundreds of high-load samples within a ride; random measurement noise therefore averages out and affects the per-log statistics far less than the raw sample-level values. Second, the degradation patterns of interest in this work correspond to large-amplitude changes in these metrics: in the synthetic ageing scenarios, equivalent and battery resistance roughly double between early life and late-life regimes, while in the real data the largest deviations from baseline amount to several standard deviations of the per-wheel reference. These amplitudes exceed the assumed instrumental uncertainty by at least one order of magnitude.

Systematic biases in voltage and current measurements, which are fixed for a given wheel, cancel in intra-wheel comparisons and only affect cross-wheel comparisons. Since `EUC_SOH` is explicitly designed to use each wheel as its own reference, and we do not attempt to compare absolute resistance levels between different models, the conclusions of this study are robust to such biases. A more detailed Monte Carlo treatment of measurement error would therefore not materially change the diagnostic conclusions for the drift magnitudes considered here; the dominant source of log-to-log variability remains the diversity of operating conditions (ambient temperature, SoC, load profile) rather than instrumental noise.

For cases where the initial state is already degraded (e.g. second-hand vehicle), a complementary absolute alarm based on

`R_pack_nominal` provides a usage-context-independent safety net.

A summary alarm table lists the logs that exceed these thresholds, together with their values and deviations from the reference. Similar Gaussian-threshold approaches are widely used for anomaly detection in battery packs and quality control, with the usual caveats regarding non-Gaussian distributions and the choice of the reference sample [10, 24, 25, 28, 35].

This vehicle-specific baseline $\mu \pm 2\sigma$ serves as the reference for alarm thresholds, ensuring that deviations are detected relative to each wheel's normal operating range rather than arbitrary fixed limits.

In addition to these statistical thresholds, `EUC_SOH` applies an absolute rule on `ELP_median`. A nominal pack resistance is estimated from the inferred number of series cells (based on maximum pack voltage) and typical cell resistance values from public data for 18650/21700 chemistries, and an alarm is raised if `ELP_median` exceeds a configurable multiple of this nominal (e.g. 1.8 $\times$) beyond a given mileage tuned so that it flags clearly abnormal but not borderline cases on the wheels considered. The goal is to flag objectively high resistances even if the local statistics have "normalized" an already degraded pack.

3.8 Trend and change detection

To characterize long-term degradation and detect regime changes, `EUC_SOH` couples the above indicators with simple trend and change-detection mechanisms. For selected metrics (e.g. `R_batt_median_25C`, `R_mosfet_hot`, `ELP_median`, `sag_95p`), we fit linear trends as a function of wheel mileage and report slopes to quantify statistically significant increases. We also apply a one-sided cumulative sum (CUSUM) change detector to these time series, using an early-life subset of logs as a reference and

monitoring the remaining logs for sustained shifts in the mean. In addition to per-metric CUSUM, EUC_SOH performs a simple regime classification at the wheel level. Using the cumulative mileage and the outputs of the change detectors on resistance and sag indicators, we label intervals of the life trajectory as "slow" or "accelerated" regimes, corresponding respectively to baseline ageing and statistically significant steps in degradation. These labels are used only for presentation purposes in the reports (e.g. annotating plots with a "Slow regime" background) and do not affect the underlying health metrics or alarm thresholds. This combination of regression-based trend analysis, Gaussian thresholds and CUSUM is intended to provide transparent, interpretable signals of degradation and abnormal behaviour, rather than a single black-box decision. In the experimental section, we illustrate how these methods highlight both gradual resistance increases over tens of thousands of kilometers and isolated, large out-of-family events on individual logs.

For CUSUM-based change detection, we use a simple, fixed parametrisation across wheels and metrics. For each metric m , the reference mean μ_{ref} and standard deviation σ_{ref} are computed on the first 30% of the lifetime in mileage, which we treat as an early-life healthy baseline. A one-sided CUSUM for upward shifts ("higher is worse") is then applied to the remaining logs, with decision level h corresponding to $\mu_{\text{ref}} + 3\sigma_{\text{ref}}$ (Fig. A1), and an intermediate line at $\mu_{\text{ref}} + 1.5\sigma_{\text{ref}}$ used for visualisation only.

This parametrisation follows standard statistical process control practice, where CUSUM thresholds at 3σ above the in-control mean provide a transparent compromise between robustness to noise and sensitivity to sustained shifts. In our case studies, the detected change points on resistance and sag metrics were stable when varying the reference fraction between 20% and 40% of the mileage span and when shifting h between $2.5\sigma_{\text{ref}}$ and $3.5\sigma_{\text{ref}}$, so we keep the same values for all wheels to favour simplicity and interpretability.

Linear trends are fitted by ordinary least squares on each metric as a function of mileage. For the indicators highlighted in Section 4 (battery resistance at 25°C, global Equivalent Load Proxy and 95th-percentile sag), the estimated slopes are positive and the corresponding p-values are well below 0.01, even when subsampling the data by a factor of two. We therefore report only the slope magnitudes in the plots, as the statistical significance of the trends is not in doubt for the datasets considered.

3.9 Aging simulator and synthetic ride logs

In addition to analysing real ride logs, we designed an ageing simulator to generate synthetic SoH trajectories and corresponding WheelLog-like files. The goal of this simulator is not to reproduce the full multi-physics coupling of electrochemical, thermal and mechanical phenomena in lithium-ion batteries or power devices, but to provide *stylized, statistically consistent* scenarios that stress-test the EUC_SOH indicators and detection mechanisms. In particular, the simulator does not implement a coupled electro-thermal Thévenin-equivalent-circuit model and does not claim to reproduce the physical mechanisms of battery ageing, MOSFET ageing, or real-world loss-of-power events. Its purpose is limited to generating controlled indicator-level what-if scenarios with known injected changes, so that the responsiveness of the monitoring, thresholding, and change-detection pipeline can be assessed. Instead, the simulator focuses on reproducing plausible long-term trends and accelerated degradation regimes (knees or

elbows) in the *observed indicators themselves* (Equivalent Load Proxy, sag, thermal proxies, current-stress measures), while preserving the marginal distributions and correlations seen in a given real wheel.

Starting from a real wheel folder, we first compute per-log health indicators and Gaussian alarm thresholds as described in Sect. 3.6 (means, standard deviations and directions "higher_is_bad" or "lower_is_bad"). These thresholds are used as a statistical prior for each metric m in the set $\{\text{ELP_median}, \text{ELP_95p}, \text{sag_95p}, \text{sag_max}, \text{v_min_strong}, \text{i_max}, \text{i_95p}, \text{i_phase_95p}, \text{i_phase_max}, \text{I_phase2_int}, \text{temp_board_max}, \text{temp_motor_max}, \text{R_batt_median}, \text{R_mosfet_hot}\}$

Given a synthetic mileage trajectory $\{k_i\}_{i=1}^N$ over several years (e.g. weekly steps of 100 km), the simulator generates a value series $m(k_i)$ for each indicator. For each metric, we use:

- The empirical mean μ_m and standard deviation σ_m from the real-wheel reference as a baseline;
- A target offset of the order of $\alpha\sigma_m$ at the end of life (with α configurable), representing the total magnitude of degradation in that metric;
- A two-stage drift model with a slow linear increase over most of the mileage span and a faster increase after a knee point at a fraction k_{knee} of the mileage (typically 90% of the span), mimicking the transition to accelerated ageing observed in real batteries [12, 13];
- Additive noise with variance smaller than σ_m^2 , to reflect intra-log variability without hiding the long-term trend.

We distinguish battery-related metrics (ELP_median, ELP_95p, R_batt_median, sag_95p, sag_max) from power-electronics-related metrics (R_mosfet_hot, temp_board_max, i_phase_95p, i_phase_max, I_phase2_int) and allow different simulation modes:

- *global*: all metrics drift with comparable magnitude;
- *batt_only*: battery metrics exhibit a strong, clearly visible drift, while MOSFET metrics remain almost flat;
- *mosfet_only*: MOSFET-related metrics drift significantly, while battery metrics change very little.

In *batt_only* mode, for instance, the final offset for battery metrics is scaled up and forced to be positive (higher resistance and sag), while MOSFET metrics are given a very small drift. For metrics where lower values are worse (e.g. v_min_strong), the drift sign is adjusted to respect the "higher_is_bad" / "lower_is_bad" convention.

Because the simulator operates at the level of the derived indicators rather than at the level of detailed electrochemical state variables, the resulting synthetic trajectories should be interpreted as *what-if* scenarios for the EUC_SOH pipeline: they answer questions of the form "if this wheel's Equivalent Load Proxy and sag doubled over 30 000 km according to a knee-like pattern, would the monitoring system detect it?", rather than providing quantitative predictions of real-world ageing paths. This abstraction is deliberate: it allows us to evaluate the sensitivity and specificity of the proposed indicators and

change detectors under controlled, high-contrast degradation patterns, without claiming high-fidelity physical ageing simulation.

The result of this process is a synthetic SoH summary table with one row per synthetic ride, containing the wheel mileage, a timestamp, and all simulated metrics. To obtain full ride logs compatible with the real analysis pipeline, we then generate WheelLog-like CSV files whose internal samples are consistent with the synthetic metrics. For each synthetic log, we:

- Choose a ride duration and number of samples, and generate realistic profiles of speed and battery current (with phase current derived as a speed-dependent multiple of battery current);
- Construct a voltage time series such that the median Equivalent Load Proxy matches the target `ELP_median` and the high-percentile sag matches `sag_95p`;
- Simulate board and motor temperature dynamics using simple first-order thermal models driven by I^2 , and rescale the resulting trajectories so that their maxima match the target `temp_board_max` and `temp_motor_max`.

This procedure yields synthetic WheelLog folders that can be processed by EUC_SOH exactly like real data. In Sect. 4, we use these synthetic datasets to create controlled scenarios of global degradation, battery-only degradation and MOSFET-only degradation, and to evaluate the sensitivity and specificity of our trend and anomaly detection mechanisms using controlled degradation scenarios with known injected faults.

4 Results

Table 2 references the characteristics of different EUCs of the dataset. They all have basic sensors that return system temperature, voltage, current, pwm, battery level, odometer. Wheels B, E and P have an additional phase current field. Wheels S and B have an additional motor temperature field.

4.1 Long-term monitoring of a wheel with stable observed indicators

We first illustrate EUC_SOH on an electric unicycle monitored from about 400 km to 8300 km of wheel mileage. While no external ground truth is available for this wheel's component health, the stability of the observed indicators provides a reference example in which no sustained alarm pattern is detected. The corresponding summary charts in Fig. 3 and Fig. 4 show resp. the evolution of `ELP_median`, `R_batt_median_25C`, `R_mosfet_hot` and `sag_95p` and the evolution of `temp_board_max`, `I_phase2_int`, `i_95p` and `i_max` as a function of wheel mileage for a set of rides covering roughly one year of use. The green bands and red danger thresholds are computed from the EUC history itself using the Gaussian reference procedure described in Sect. 3.6.

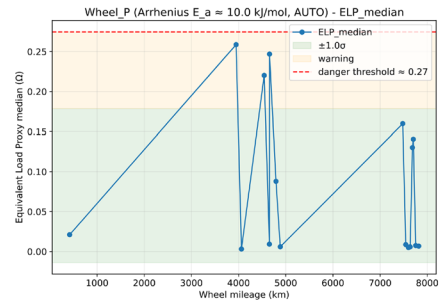
For this wheel, all resistance-based indicators remain within or close to the green bands over the observation window.

`ELP_median` and `R_batt_median_25C` exhibit only modest fluctuations around their reference values, without any clear upward trend or approach to the danger thresholds. The MOSFET-related resistance `R_mosfet_hot` remains close to its nominal range, and the voltage sag metrics `sag_95p` and `sag_max` stay well below their danger thresholds, indicating stable ohmic behaviour of the energy and power stages. Battery

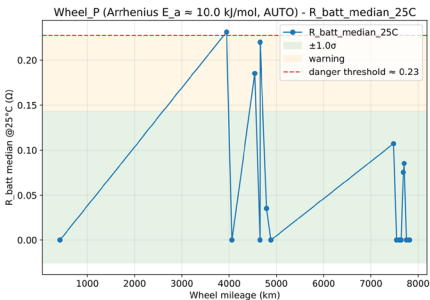
Table 2 Summary of wheels and telemetry characteristics used in the detailed case studies. Manufacturer and model names are anonymized. After applying data-quality and signal-availability criteria, 90 rides from eight wheels were retained for the static-versus-differential comparison. The detailed case studies use 43 logs from three wheels

Wheel	Pack V (nominal)	Cells (approx.)	Energy (approx.)	Sampling (Hz)	Additional channels
Wheel B	134.4 V	32s4p (21700)	~2.4 kWh	5–10	Phase current, motor temp
Wheel E	134.4 V	32s4p (21700)	~2.4 kWh	5–10	Phase current
Wheel I	235.2 V	56s4p (21700)	~4.2 kWh	5–10	
Wheel M	84 V	20s2p (21700)	~0.7 kWh	5–10	
Wheel P	134.4 V	32s8p (21700)	~4.8 kWh	5–10	Phase current
Wheel S	126 V	30s4p (21700)	~2.2 kWh	5–10	Motor temp
Wheel V	84 V	20s2p (21700)	~0.7 kWh	5–10	
Wheel Z	58.8 V	14s3p (18650)	~1.0 kWh	1–5	

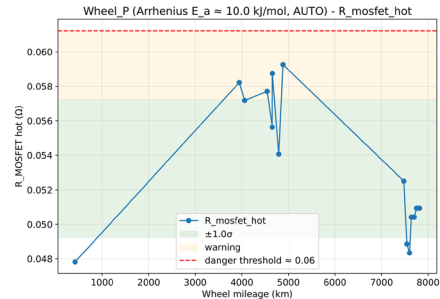
Wheel	# rides	Mileage range (km)	Motor (W)	Usage	Firmware
Wheel B	15	800–2,200	3,500	street	N/A
Wheel E	7	0–500	3,500	off-road	N/A
Wheel I	5	0–1,200	6,000	street	N/A
Wheel M	11	150–900	1,000	urban	N/A
Wheel P	16	0–8,000	4,500	street	N/A
Wheel S	17	0–2,500	4,000	mixed	N/A
Wheel V	7	0–700	1,000	urban	N/A
Wheel Z	12	0–700	1,800	street	N/A



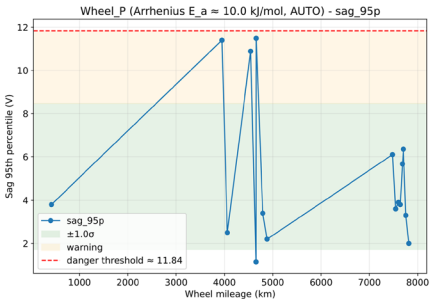
(a) Median equivalent pack resistance ELP_{median} vs wheel mileage for Wheel P (intra-wheel indicator combining battery and power stage).



(b) Battery-dominated pack resistance $R_{batt,median@25^{\circ}C}$ vs wheel mileage for Wheel P (intra-wheel indicator normalised to $25^{\circ}C$).

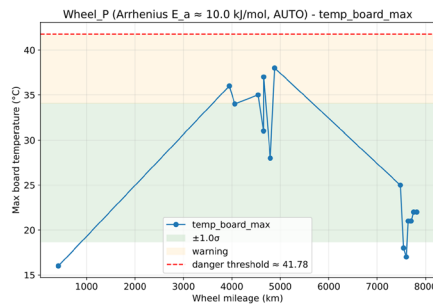


(c) MOSFET + wiring resistance proxy $R_{mosfet,hot}$ vs wheel mileage for Wheel P (estimated from board-temperature proxy).

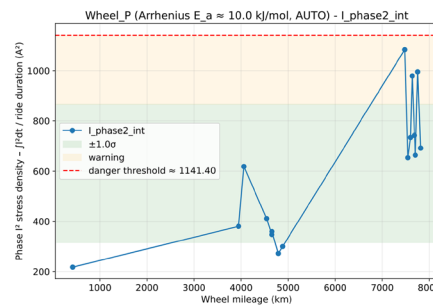


(d) 95th-percentile voltage sag $sag_{95\%}$ vs wheel mileage for Wheel P under high-load conditions.

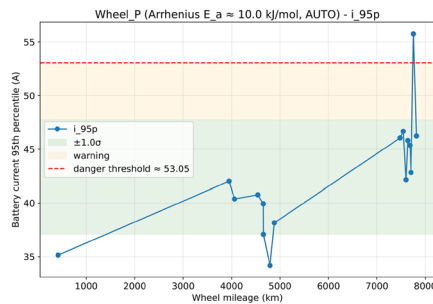
Fig. 3 EUC_SOH pack-level health indicators for Wheel P over its monitored lifetime. All curves are intra-wheel, using this wheel as its own reference, and combine the effects of battery, power stage and wiring under the data and thermal-proxy limitations described in Sect. 3.6



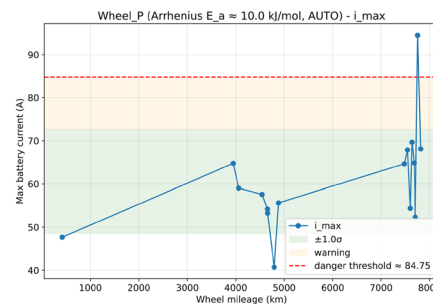
(a) Maximum board/controller temperature **temp_board_max** vs wheel mileage for Wheel P (proxy for power-stage thermal stress).



(b) Phase-current stress density **I_phase2_int** vs wheel mileage for Wheel P (mean I^2 over ride duration, proxy for sustained electrothermal stress).



(c) Battery current 95th percentile **i_95p** vs wheel mileage for Wheel P.



(d) Maximum battery current **i_max** vs wheel mileage for Wheel P.

Fig. 4 Thermal and current-stress indicators for Wheel P over its monitored lifetime (intra-wheel view). These indicators characterise the typical and extreme electrical and thermal loading experienced by the power stage and are used to contextualise the resistance and sag trends in Fig. 3

Table 3 Alarmed logs for Wheel P: mileage and compact alarm reasons

Wheel km	Alarm reasons (compact)
418.8	high ELP _{med} , high phase $I_{95\%}$
3946.3	huge ELP (med, 95%), sag↑ (95%, max), low $V_{\min}$, high R_{batt}
4542.0	huge ELP (med, 95%), sag↑ (95%, max), low $V_{\min}$, high R_{batt}
4655.6	huge ELP (med, 95%), sag↑ (95%, max), low $V_{\min}$, high R_{batt}
4792.6	$R_{\text{eq,med}}\uparrow$, $R_{\text{batt}}\uparrow$ (moderate)
7479.0	ELP↑ (med, 95%), sag↑ (95%, max), $R_{\text{batt}}\uparrow$
7685.0	ELP↑ (med, 95%), sag↑ 95%, $R_{\text{batt}}\uparrow$
7703.8	ELP↑ (med, 95%), sag↑ (95%, max), $R_{\text{batt}}\uparrow$

current percentiles and maximums, as well as board and motor temperatures, also remain within expected ranges given the usage patterns, with no alarming excursions.

This case illustrates a wheel with stable observed indicators over more than 7,800 km of additional mileage, providing a compact intra-wheel view of battery and power-stage-related behaviour. For such a wheel, the alarm table 3 remains sparse and mostly highlights isolated rides with slightly higher stress, which can be interpreted as normal variation rather than signs of degradation.

Yet, the last ride on the monitored wheel P illustrates the interest of monitoring those machines for safety purposes. For, during that run, the rider reported no obvious symptoms at the vehicle level (no tilt-back, no audible alarms, no over-temperature warnings), and the ride felt subjectively “normal”. In the EUC_SOH analysis, however, the corresponding log sits near the top of the wheel’s historical distribution for electrical

stress: both the 95th-percentile and maximum battery currents exceed the wheel's internal danger thresholds, while phase-current stress as measured by the $i_{\text{phase_95p}}$ and $i_{\text{phase_max}}$ in Fig. 5 is among the highest values observed for this wheel, even though resistance and sag metrics remain within their nominal ranges. In other words, the session was significantly more demanding on the power stage than typical rides, without any immediate feedback from the wheel itself. This type of situation motivates the use of long-term, log-based health and usage monitoring: it is precisely when the vehicle still behaves “normally” that riders are most likely to accumulate high-stress runs without realizing how hard they are driving the hardware.

4.2 Simulated progressive indicator drift consistent with degradation on wheel P (Wheel_P_bad)

To evaluate EUC_SOH's sensitivity, we generate synthetic ageing scenarios where degradation is artificially injected (e.g., doubling R_{batt} after a knee point). These scenarios serve as controlled test cases with known degradation patterns, allowing us to verify that the system detects the expected trends and outliers.

We next consider a synthetic ageing scenario derived from the same EUC wheel P. Using the aging simulator described in Sect. 3.8, we generate a synthetic SoH trajectory over approximately 30–35 000 km. This data is created with controlled, progressive indicator drift consistent with degradation affecting both battery and MOSFET-related indicators (mode *global*). The corresponding Wheel_P_bad report shows clear increasing trends in $R_{\text{batt_median_25C}}$, $R_{\text{mosfet_hot}}$, ELP_{median} and sag_95p as mileage increases, as shown in Fig. 6.

For Wheel P, histograms of ELP_{median} (Fig.) show that the synthetic trajectory remains within the empirical range of the real wheel in the early-life region, while its late-life tail extends several standard deviations above the reference, as intended by the degradation model.

Linear trend fits yield positive slopes on all four indicators, with $R_{\text{batt_median_25C}}$ showing a slope on the order of $10^{-2} \Omega/1000 \text{ km}$, which reflects a sustained increase in both battery and global Equivalent Load Proxy.

Change-point detection on $R_{\text{batt_median_25C}}$,

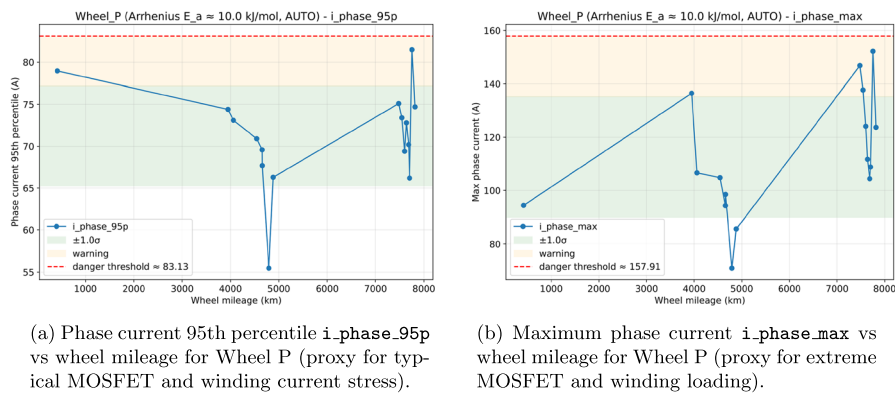
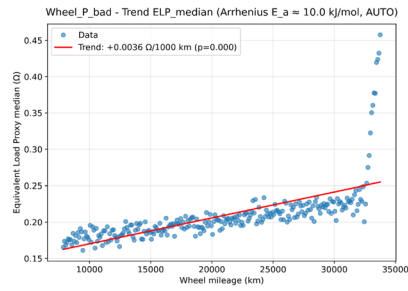
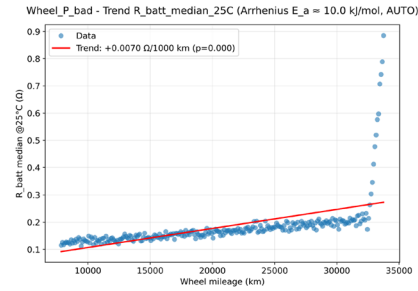


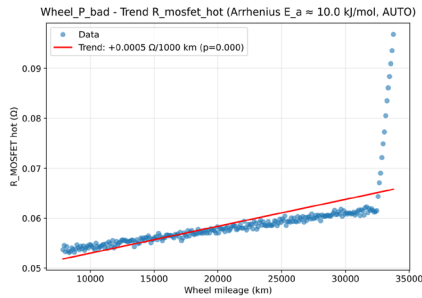
Fig. 5 Phase-current-based stress indicators for Wheel P over its monitored lifetime (intra-wheel view). These quantities do not model junction temperature explicitly, but act as empirical proxies for the electrical loading experienced by the MOSFET bridge and motor windings in the absence of device-level thermal measurements



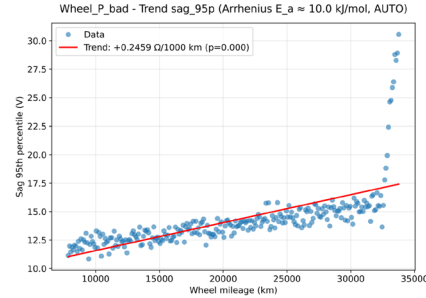
(a) Median equivalent pack resistance ELP_{median} vs wheel mileage for the synthetic $Wheel_P_bad$ trajectory (stylized ageing scenario).



(b) Battery-dominated pack resistance $R_{batt,median@25^{\circ}C}$ vs wheel mileage for $Wheel_P_bad$.



(c) MOSFET + wiring resistance proxy $R_{mosfet,hot}$ vs wheel mileage for $Wheel_P_bad$.



(d) 95th-percentile voltage sag $sag_{95\%}$ vs wheel mileage for $Wheel_P_bad$ under high-load conditions.

Fig. 6 stylized synthetic ageing scenario for Wheel P ($Wheel_P_bad$). The synthetic indicators are constructed by amplifying late-life trends in the real wheel's pack-level metrics to emulate a knee-like degradation pattern, while preserving early-life behaviour. These trajectories are used to stress-test the EUC_SOH indicators and change detectors rather than to predict a specific physical ageing path

Table 4 Synthetic Wheel P: Selected alarmed logs at early and late mileage (compact alarm reasons)

Wheel km	Alarm reasons (compact)
10 455	Motor T_{max} slightly above limit
11 955	High phase I^2 dose
12 655–13 255	Sparse motor T_{max} , phase $I_{95\%}$ alarms
...	...
33 255	Huge ELP (med, 95%), sag↑ (95%, max), low V_{min} , high R_{batt} , high board T_{max}
33 355	Huge ELP (med, 95%), sag↑ (95%, max), low V_{min} , high R_{batt} , high board T_{max}
33 455–33 755	Same pattern, with further increase in R_{batt} , R_{mosfet} , sag and board temperature

R_{mosfet_hot} , ELP_{median} and sag_{95p} shows clear late-life regime shifts, with the cumulative sums crossing the decision thresholds consistently with the accelerated degradation imposed by the simulator, cf. Appendix .

In parallel, Table 4 shows that in the last 5 000 km of the synthetic trajectory, most rides are flagged as *danger* on

ELP_{median} , $R_{batt_median_25C}$ and sag_{95p} . Also v_{min_strong} approaches the low-voltage danger threshold, reflecting a substantial loss of electrical margin compared to the real Wheel P reference.

Because the synthetic trajectories are constructed so that the final offsets correspond to several standard deviations of the real Wheel P reference, we can treat this scenario as a controlled positive test for the detection mechanisms. The results indicate that EUC_SOH correctly identifies long-term drift in both battery and MOSFET-related indicators,

Table 5 Mileage (km) at first trigger of each alarm type across simulated degraded wheels. CUSUM: first CUSUM change-point per wheel. Stat. threshold: first multi-indicator $\mu+2\sigma$ alarm. Knee: first slope inflection detected. EOL: first absolute threshold alarm. “—” = not triggered. Last column: CUSUM lead time over the statistical threshold (positive = earlier detection by CUSUM)

Wheel	Total simulated km	CUSUM (km)	Stat. threshold (km)	Knee (km)	EOL (km)
P	33,754	10,154	11,854	32,754	—
E	26,327	8,027	5,727	25,327	—
B	27,829	8,529	10,029	26,729	27,529
M	26,891	8,191	11,691	25,891	—
S	26,978	8,178	7,678	25,978	—
V	26,697	8,197	11,697	25,597	—
Z	26,742	8,042	8,042	25,742	—

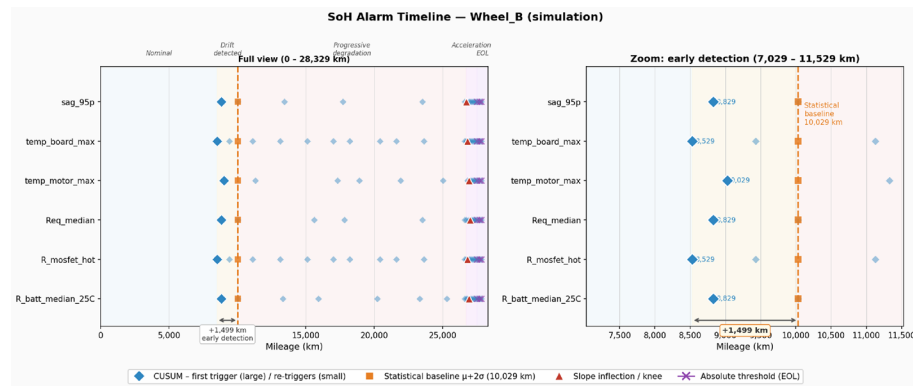


Fig. 7 SoH alarm timeline for simulated Wheel B. Left: full mileage view (0–28,000 km) showing the four alarm types (CUSUM change-point, statistical threshold $\mu+2\sigma$, slope inflection/knee, absolute EOL threshold) overlaid on the degradation timeline per monitored metric. Right: zoom on the early-detection window (7,000–11,500 km), highlighting the +1,500 km lead of the CUSUM detector over the statistical threshold. Background colours indicate degradation phases: nominal (blue), drift detected (yellow), progressive indicator drift consistent with degradation (pink), acceleration/knee (light purple), EOL (purple)

in this synthetic scenario. It raises alarms when resistance and sag enter clearly abnormal regimes, and provides interpretable diagnostics in terms of gradual loss of margin.

4.2.1 Comparison with baseline for early detection and detection accuracy

Table 5 summarizes the mileage at which each alarm type first triggers across all simulated degraded wheels. Across the controlled synthetic scenarios, the CUSUM detector provides earlier or comparable detection relative to the per-metric statistical threshold for most simulated wheels, while the statistical threshold is reached first in the remaining cases. For Wheel P, the CUSUM fires at 10,154 km, approximately 1,700 km before the first multi-indicator statistical alarm at 11,854 km. The slope inflection (knee) is detected in the 25,000–33,000 km range, consistent with the accelerated degradation phase of the simulator, while absolute EOL alarms are only triggered for Wheel B at 27,529 km. Figure 7 illustrates this timeline for Wheel B.

Across the degraded wheels, the CUSUM detector either provides a clear early-warning compared to the per-metric $\mu+2\sigma$ statistical threshold (e.g. Wheels P, B, M and V) or matches it (Wheel Z). For Wheels E and S, the first statistical threshold alarm occurs slightly earlier than the first CUSUM change-point, reflecting the fact that the fixed CUSUM parametrisation used in this work is somewhat more conservative on those trajectories. Overall, these results confirm that, in the controlled synthetic scenarios,

CUSUM offers earlier or comparable detection while maintaining a uniform, interpretable parameter setting across all wheels.

4.3 Detection of a single major deviation on a long-life wheel (Wheel S)

Finally, we examine a synthetic scenario based on EUC named wheel S with a lifetime of real data extended synthetically to about 26 000–27 000 km. For most of the mileage range, the Wheel_S_bad report shows a slow, regular evolution of ELP_median , $R_batt_median_25C$, R_mosfet_hot and sag metrics, all remaining within the green band or slightly above, consistent with a “slow ageing” regime. In this regime, only a limited number of logs cross the danger thresholds, and the alarm table remains relatively sparse.

Near the end of the mileage range, a single synthetic log (e.g. around 26 000–27 000 km) is constructed to exhibit a large, abrupt degradation in a single indicator: R_mosfet_hot jumps to values well above its danger thresholds. As shown in Fig. 8 Board temperature also reaches unusually high values for this wheel, reflecting severe stress on the power stage.

EUC_SOH flags this log as an out-of-family event across multiple metrics: it appears as an isolated red point in the resistance and temperature plots, well separated from the cloud of previous logs, and is prominently listed in the alarm table with large deviations from the reference. Importantly, this detection occurs even though global statistics (e.g. overall mean and standard deviation across the full life) remain dominated by the long period of normal operation. The method therefore proves capable of highlighting a single major deviation at the end of an otherwise regular trajectory, which is precisely the type of event that could hypothetically precede or coincide with a dangerous loss-of-power incident.

Together, these three cases show that EUC_SOH can (i) confirm stable operation for a wheel with stable observed indicators, (ii) detect and quantify progressive indicator drift consistent with degradation over tens of thousands of kilometers, and (iii) isolate rare but severe out-of-family events on individual logs in a long-life wheel. In the discussion, we analyse how such information could be used to support maintenance decisions and to better understand safety margins for EUC riders.

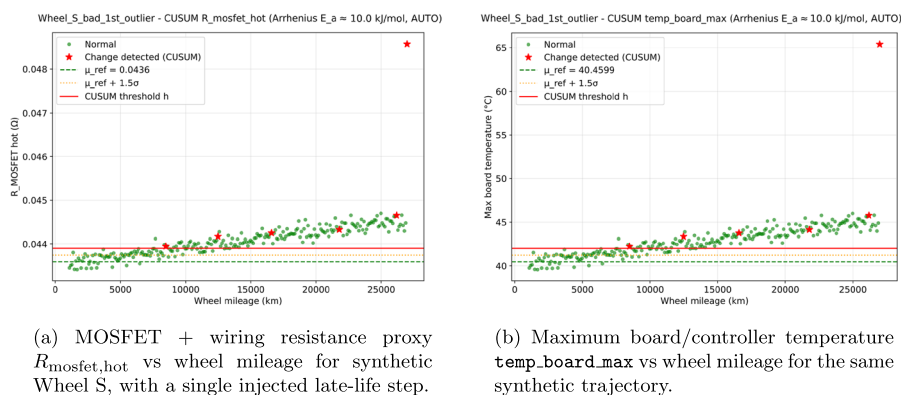


Fig. 8 Detection of a single major event on linked indicators for synthetic Wheel S. A stylized step change is injected in the MOSFET-dominated resistance proxy and associated board temperature to emulate a sudden degradation or partial failure. EUC_SOH is used here to illustrate how such out-of-family events would appear in the pack-level indicators under the constraints of EUC telemetry

Overall, the present study covers three real wheels with monitored mileage ranging from about 900 km to 7 800 km and a total of 43 ride logs, complemented by synthetic trajectories extending individual wheels up to 25 000–35 000 km. While this represents only a small sample of the EUC population, it already spans different usage profiles, with some wheels used mostly for moderate commuting and others exhibiting frequent high-current, high-sag runs.

5 Discussion

5.1 Benchmarking considerations

While classical PHM approaches often compare against fixed baselines (e.g., manufacturer specifications), such an approach is inapplicable to EUCs for two reasons:

1. Lack of standardized data: EUC manufacturers do not disclose cell-level parameters or nominal pack resistances, making absolute thresholds unreliable.
2. Usage variability: A single EUC may operate under wildly different conditions (e.g., urban commuting vs. off-road), causing natural variations in R_{eq} that would trigger false alarms with fixed thresholds.

Our adaptive baseline ($\mu \pm 2\sigma$ per wheel) addresses both issues by focusing on relative changes within a vehicle's own history, as demonstrated in Sects. 4.2, 4.3 where it successfully detects synthetic degradation trends that a fixed threshold would miss until late stages, thus illustrating its potential advantage over the tested fixed-threshold policies under the simulated scenarios.

5.2 Quantitative comparison against fixed-threshold baselines

To quantitatively benchmark EUC_SOH against simpler detection strategies, we compare three alarm policies on the ELP_median indicator across all simulated wheels: (i) the adaptive per-wheel statistical threshold ($\mu+2\sigma$) used by EUC_SOH, (ii) a fixed global threshold ($\mu_{global}+3\sigma_{global} = 0.395 \Omega$) computed from the first 30% of real-wheel data pooled across all wheels, and (iii) an absolute EOL threshold set at $2 \times R_{pack,nominal}$ per wheel.

As shown in Table 6, the fixed global threshold ($\mu+3\sigma$) triggers between 0 and 13 alarms per wheel, missing 127–148 alarms detected by EUC_SOH. The absolute EOL threshold triggers zero alarms on all wheels, as ELP_median never reaches $2 \times R_{pack,nominal}$ during normal operation. This confirms that fixed baselines are either too conservative (missing early degradation) or too permissive (triggering only at end-of-life), and that adaptive per-vehicle baselining is necessary given the high inter-wheel variability in nominal resistance (μ_{wheel} ranges from 0.115 Ω to 0.237 Ω across wheels).

5.3 Implications for EUC safety and usage

The results suggest that long-term monitoring of pack-level resistance, sag and thermal indicators can provide early, interpretable signals of degradation in electric unicycles, even when only user-accessible telemetry is available. On a wheel with stable observed indicators, EUC_SOH confirms the stability of key indicators over more than 7800 km of additional mileage, while in synthetic ageing scenarios it detects both gradual resistance increases and single, large out-of-family events after tens of thousands of kilometers. In practice, such information could be used to trigger precautionary checks, derating or

Table 6 Comparison of EUC_SOH adaptive threshold vs. fixed baselines on ELP_median across simulated wheels, $\mu_{global} + 3\sigma_{global} = 0.395 \Omega$ (computed from pooled early-life real data). EOL threshold: $2 \times R_{pack,nominal}$ per wheel, $\Delta = EUC_SOH$ alarms minus fixed-threshold alarms (positive = more alarms detected by EUC_SOH)

Wheel	$\mu_w(\Omega)$	$\sigma_w(\Omega)$	$\tau_{EUC_SOH}(\Omega)$	$\tau_{\mu+3\sigma}(\Omega)$	$\tau_{EOL}(\Omega)$	#alarms	$\Delta vs \mu+3\sigma$
P	0.187	0.010	0.207	0.395	0.896	144	+140
E	0.160	0.013	0.185	0.395	0.840	132	+127
B	0.237	0.023	0.283	0.395	0.896	152	+139
M	0.115	0.019	0.154	0.395	0.880	138	+132
S	0.136	0.010	0.155	0.395	0.840	150	+148
V	0.130	0.014	0.159	0.395	0.880	139	+134
Z	0.146	0.008	0.162	0.395	0.704	148	+148

recommended maintenance when a wheel approaches abnormal regimes in battery or power electronics health, thereby possibly reducing the likelihood of operating a severely degraded vehicle at high load.

It is important to emphasize that EUC_SOH is not a crash-prediction system. Current micromobility safety statistics do not distinguish EUC hardware failures from crashes caused by collisions or loss-of-control [1–3], and we do not claim that any specific incident could have been prevented by the proposed monitoring alone. Rather, EUC_SOH provides a way to structure and interpret the health information contained in ride logs, in a form that is comparable to HUMS indicators used in other vehicle classes. In the longer term, systematically logging such health indicators alongside incident reports could help build the data needed to understand if and how component degradation contributes to loss-of-power risk in real fleets.

Likewise, the alarm bands used in this work are screening tools rather than safety-certified decision limits. Their purpose is to make unusual intra-wheel behaviour visible and interpretable, not to prescribe mandatory operating restrictions or to certify a vehicle as safe or unsafe.

From a SoH perspective, the indicators used in EUC SOH can be read as a compact description of how “hard” it has become for the vehicle to push current through its powertrain. As long as ELP_{median} , battery-dominated resistance and sag metrics fluctuate within a narrow band, the wheel behaves like a healthy pack: voltage dips remain modest under load and the thermal envelope stays well below the firmware limits. When these quantities drift upwards and the minimum strong-load voltage approaches the cut-off threshold, the practical state of health decreases even if the pack still charges to its nominal voltage, because less power can be delivered safely in demanding situations. In this sense, EUC SOH does not attempt to predict an exact remaining lifespan, but to track when a wheel transitions from “normal operation” to regimes where reduced performance, derating or preventive maintenance become reasonable safety decisions.

5.4 Positioning relative to existing HUMS and EV analytics

From an architectural standpoint, EUC_SOH mirrors key elements of Health and Usage Monitoring Systems: on-board sensing, ride-level feature extraction, statistical baselining, trend and change detection [4–7]. However, it operates under stricter constraints, relying solely on the telemetry exposed to end-users and without access to internal BMS cell-level measurements or high-frequency switching waveforms. In that sense, it is closer to data-driven EV battery analytics using telematics than to intrusive lab instrumentation [8–11, 25, 28, 29].

The proposed framework could be extended to other micromobility devices that expose similar telemetry (e-scooters, e-bikes), but electric unicycles constitute an interesting extreme case: their dynamic stability makes them particularly sensitive to loss-of-power events, while their high performance and relatively open protocols provide rich data streams. EUC_SOH therefore offers a concrete example of how HUMS concepts can be adapted to personal micromobility, and highlights the opportunity for manufacturers and app developers to expose and exploit health indicators beyond simple dashboards.

Beyond the three detailed case studies, the ageing simulator was applied systematically to all wheels with sufficient historical data. For each wheel, we generated three synthetic

trajectories (battery-only degradation, MOSFET-only degradation and global degradation) using the procedure of Sect. 3.8, and processed the resulting logs through the same EUC_SOH pipeline. In addition, for Wheel S we created a dedicated single outlier scenario to emulate an abrupt late-life MOSFET degradation event. These experiments confirm that the combination of resistance, sag and thermal indicators, together with the Gaussian thresholds and CUSUM-like change detectors, behaves consistently across multiple wheels and ageing modes, while remaining interpretable in terms of battery versus power-stage stress.

Although EUC_SOH is tailored to electric unicycles, most of the underlying ideas are generic to electric powertrains, while a few aspects are EUC-specific.

On the generic side, the method relies on pack-level voltage and current measurements, simple Joule-loss arguments and statistical baselining of resistance, sag, current and temperature indicators over long-term usage. These ingredients are directly applicable to other PEVs and EVs that expose similar telemetry.

On the EUC-specific side, EUC_SOH assumes wheels where only pack-level signals and coarse temperature proxies are available to end-users, with no access to BMS cell-level data or high-frequency inverter waveforms. The idle-voltage extraction based on quasi-idle plateaus and the use of wheel-level odometer mileage as the main ageing axis are design choices that reflect this constraint and the typical logging capabilities of EUC companion apps. Extending the approach to other platforms equipped with richer diagnostics (e.g. per-cell voltages, internal impedance estimates) would mainly require replacing the idle-voltage proxy and resistance decomposition with more direct measurements, while keeping the same high-level workflow (per-ride feature extraction, statistical reference building, trend and change detection).

Relationship between ELP and brownout risk. In the context of electric unicycles, a brownout designates a voltage collapse severe enough to cause the controller firmware to reduce motor torque, trigger a low-voltage cutoff, or lose balance control, even though the battery is not fully discharged. This is distinct from a full power-off event: the pack retains energy, but the combined effect of degraded internal resistance, high current demand and low state of charge causes the terminal voltage to drop below firmware protection thresholds. Because ELP directly quantifies how much voltage is lost across the pack for a given current, a sustained increase in `ELP_median` is an interpretable indicator of reduced voltage margin and potential brownout risk: for the same current demand and state of charge, a wheel with a high ELP will reach protection thresholds earlier, with less warning, and under conditions (moderate speed, not-quite-empty pack) where the rider would not normally expect a loss of propulsion. The `sag_95p` and `v_min_strong` indicators further translate this risk into absolute voltage margins, making them directly interpretable in terms of firmware cut-off proximity. This connection between ELP growth and brownout probability is the primary safety motivation for the vehicle-level monitoring proposed in this work.

5.5 Practical implications for manufacturers

Beyond health monitoring itself, the present work also indirectly highlights the role of community-developed logging tools. The ride logs analyzed here exist only because independent developers have reverse-engineered proprietary Bluetooth Low Energy (BLE) frames exposed by EUC manufacturers and implemented third-party applications

capable of recording pack-level telemetry over long periods of time. In practice, BLE traffic is broadcast in the clear and will be decoded sooner or later; manufacturers would therefore benefit from (i) converging towards more standardised telemetry payloads across models and brands, and (ii) publishing stable BLE interface contracts so that user-facing applications and diagnostics tools can be implemented without repeated reverse-engineering efforts. Such openness would reduce duplicated work, improve data quality and, ultimately, make it easier to build safety-relevant health and usage monitoring tools on top of EUC telemetry.

5.6 Limitations and future work

Several limitations must be acknowledged. First, the separation between battery and MOSFET contributions relies on simplified thermal and electrical models, and on user-supplied parameters such as $R_{ds,on}$ and its temperature coefficient; errors in these inputs can bias the estimated battery resistance. Second, the statistical baselining assumes that a sufficient number of early-life logs represent a healthy state, and uses Gaussian thresholds that may be suboptimal for skewed or heavy-tailed distributions [10, 28, 35]. While Gaussian bands are only an approximate model of the indicator distributions, the appendix example in Fig. shows that, for the wheels studied, even large late-life deviations form a small enough fraction of the data that they do not destabilize the reference mean and standard deviation. Third, environmental and usage factors (ambient temperature, rider mass, terrain, tire pressure) can significantly affect sag, resistance and temperatures, potentially leading to false positives or false negatives if not accounted for. Moreover, the use of a local $V_{idle,local}$ profile built from quasi-idle plateaus and, when available, voltage-based SoC information provides only an approximation of the true open-circuit voltage. Brief pauses in real-world riding do not guarantee complete electrochemical relaxation, and the estimate does not explicitly model voltage hysteresis, diffusion polarization, or relaxation dynamics. Consequently, the resulting sag and ELP quantities are best interpreted as comparative indicators within a given wheel rather than as precise absolute measurements of electrochemical impedance.

For one additional wheel (Wheel M), the battery pack was replaced during the observation window at around 750 km, leading to a discontinuity in resistance and sag indicators rather than a smooth ageing trajectory. EUC_SOH reflects this as an abrupt step in `ELP_median` and `R_batt_median_25C`, followed by a new stable regime. In this case, the pack was swapped for higher-performance cells: `i_95p` reaches the highest values observed for this wheel while `sag_95p` attains its lowest values, see Appendix . This suggests that using high-output cells can significantly improve electrical margins for EUCs. While we do not analyse this case in detail here, it illustrates that pack replacement appears in the health indicators as a reset event rather than progressive indicator drift consistent with degradation and should be handled explicitly in long-term fleet studies.

The temperature normalization of battery resistance relies on board/controller temperature as a proxy. All temperature-dependent quantities (`R_batt_median_25C`, `R_mosfet_hot`, and the Arrhenius correction) use `temp_board_max` as the sole thermal observable. This quantity is a PCB-level measurement and does not represent cell junction or MOSFET junction temperatures. The corrections are therefore primarily intended to improve the comparability of logs from the same wheel across sessions with

different ambient and thermal conditions; they are not accurate enough to support precise cross-wheel or cross-platform comparisons without additional thermal calibration.

Beyond these methodological aspects, several avenues for improvement and deployment can be envisioned. Future work will address the above limitations by (i) incorporating explicit conditioning on temperature, current and speed ranges in the definition of comparable operating points; (ii) exploring more robust statistical models for baseline and anomaly detection; and (iii) integrating additional information when available, such as periodic cell-level snapshots from smart BMSs or external measurements. On the application side, we plan to develop a mobile or embedded implementation of EUC_SOH that can provide near-real-time feedback to riders, and to design an anonymized incident and health indicator registry for EUCs and other EDPMs. Such a registry would enable the community to relate concrete incidents to prior health trajectories, and to evaluate systematically whether certain degradation patterns are associated with increased loss-of-power risk.

Usage profiles remain an important confounding factor: two wheels with healthy hardware but very different riding styles (e.g. conservative commuting versus aggressive off-road use) can exhibit markedly different distributions of current and I^2 -dose indicators while sharing similar resistance and sag baselines. In the present work we partly mitigate this by focusing on within-wheel statistics and by using each wheel as its own reference, but a more systematic treatment of usage heterogeneity across wheels is left for future work.

In the longer term, similar diagnostics could be integrated directly into commercial EUCs. At present, health monitoring for electric unicycles is entirely telemetry-driven and relies on external logging applications and offline post-processing of CSV files. From a system design perspective, manufacturers could embed similar diagnostics directly in the vehicle firmware, at the cost of additional memory and computation resources. A first option would be to integrate additional non-volatile memory and continuously log raw telemetry over a sliding time window, then periodically recompute SoH metrics such as Equivalent Load Proxy, voltage sag, and phase-current stress integrals on board. A more resource-efficient option would be to aggregate per-ride statistics in real time (e.g. median Equivalent Load Proxy under high load, voltage sag percentiles, $\int I_{\text{phase}}^2 dt$, maximum board and motor temperatures) and only store these compact summaries in flash at the end of each ride, so that SoH becomes an embedded property of the vehicle rather than an ex-situ analysis. Beyond passive monitoring, these embedded SoH metrics could directly drive internal safety policies: for each ride summary, the firmware can associate quantitative health indicators with discrete alarm levels and, on the next power-up, decide whether to authorize normal operation, restrict performance, or refuse to start altogether if recent usage exceeded critical limits, while presenting explicit, interpretable messages to the rider instead of opaque generic fault codes.

From a computational standpoint, the EUC SOH offline analysis pipeline is lightweight. On a commodity laptop (8-cores CPU @ 3 GHz, 16 GB RAM, SSD), processing the full dataset used in this study (8 wheels, 90 rides, + 5838 simulated rides covering

tens of thousands of km) takes about 190 s end-to-end, including feature extraction, per-wheel baselines, change detection and report generation. This runtime is dominated by reading CSV files and computing per-ride percentiles; the statistical thresholds and CUSUM detectors are linear in the number of rides and negligible compared to I/O. As a result, the same algorithms could be deployed on resource-constrained hardware by aggregating per-ride summaries on-board and offloading only compact health indicators instead of full logs.

Algorithmically, all steps are $O(N)$ in the number of samples or rides; no matrix inversions or heavy optimization are required.

While EUC_SOH has been developed and evaluated on electric unicycles, the design choices it embodies: pack-level indicators built from end-user telemetry, intra-vehicle baselining, simple but interpretable change detection, and stylized ageing scenarios for method validation are generic. The same pattern could be applied to other personal electric vehicles that provide access to pack voltage, current, temperature and trip data, with appropriate reparameterization of thresholds and stress metrics. In that sense, EUCs can be seen as an early testbed for micromobility vehicle-level HUMS, rather than an isolated niche.

While ELP provides a robust vehicle-level indicator, it is not a direct measurement of battery internal resistance. Future work could validate our approach against ground truth methods (e.g., DCIR or HPPC tests in controlled environments) to quantify absolute accuracy. Additionally, higher-frequency telemetry (e.g., > 100 Hz) or designed pulse tests could enable differential methods, as used in automotive BMS [33].

While EUC_SOH focuses on electrical degradation, EUCs can fail due to:

- Mechanical: Bearing wear, structural fatigue (detectable via vibration sensors, not covered here).
- Electronic: BMS faults, sensor drift (require cell-level data).
- Thermal: Overheating in motor windings (partially captured by `temp_motor_max`).

Future work could integrate multi-sensor fusion for comprehensive PHM.

6 Conclusion

We have presented EUC SOH, a vehicle-level health and usage monitoring system for electric unicycles based on user-accessible ride logs. Using approximately 15,800 km of real ride logs across eight wheels, complemented by stylized synthetic trajectories extending to tens of thousands of simulated kilometres, EUC SOH extracts interpretable indicators of Equivalent Load Proxy, battery-attributed resistance, MOSFET-attributed resistance, voltage sag, current and temperature. Combined with vehicle-specific statistical baselining, trend analysis and change detection, these indicators provide a framework for characterizing stable intra-wheel behaviour, detecting indicator drift, and isolating out-of-family events. Although it does not replace formal safety certification, laboratory-grade SoH testing or crash analysis, the proposed framework demonstrates that EUCs can be monitored using structured vehicle-level health indicators and provides a foundation for bringing HUMS-like practices into the micromobility domain.

Appendix A wheel P

Synthetic data consistency

See Figure 9.

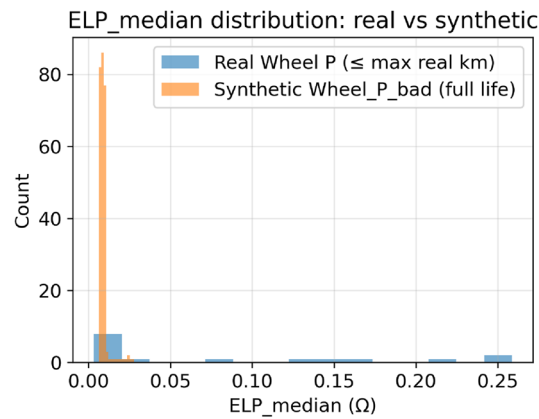


Fig. 9 Distribution of median Equivalent Load Proxy ELP_{median} for the real Wheel P (blue, up to its maximum observed mileage) and the synthetic Wheel_P_bad trajectory (orange, full simulated lifetime). The synthetic data preserve the early-life range of the real wheel while adding a late-life tail of higher resistances corresponding to the controlled degradation scenario

Gaussian stability under large deviations

See Figure 10

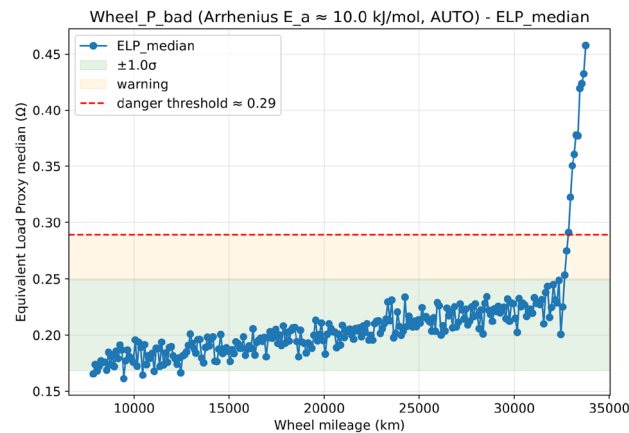
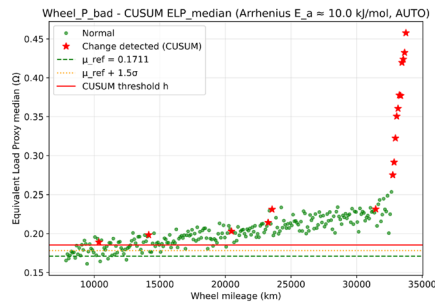


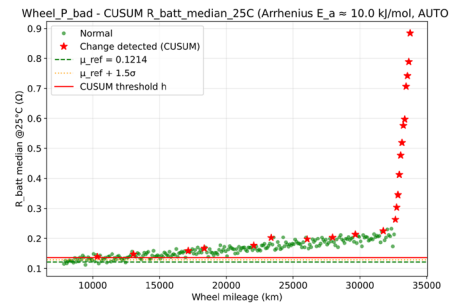
Fig. 10 Median equivalent pack resistance ELP_{median} for Wheel_P_bad with per-wheel Gaussian bands ($\mu \pm 1\sigma$ and $\mu + 2\sigma$) built from the full history. The late-life synthetic degradation produces strong out-of-family deviations, but these points form a small fraction of the dataset and have limited influence on the estimated mean and standard deviation, illustrating the robustness of the Gaussian reference to a modest number of large deviations

CUSUM detection

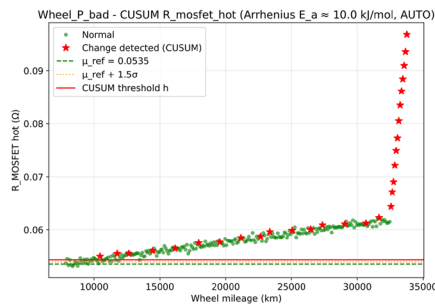
See Figures 11 and 12



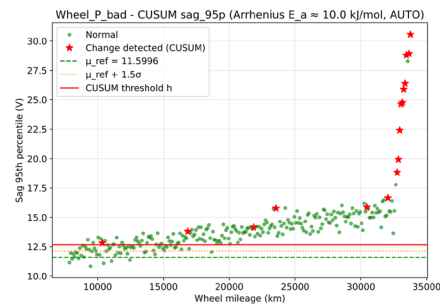
(a) Median equivalent pack resistance ELP_{median} with CUSUM-based change detection for Wheel_P_bad.



(b) Battery-dominated pack resistance $R_{batt,median@25^{\circ}C}$ with CUSUM-based change detection for Wheel_P_bad.

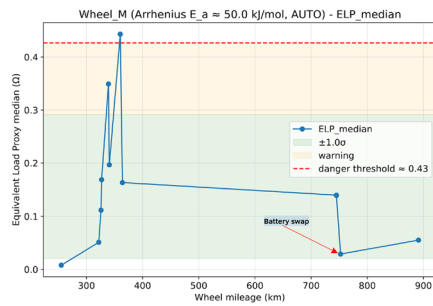


(c) MOSFET + wiring resistance proxy $R_{mosfet,hot}$ with CUSUM-based change detection for Wheel_P_bad.

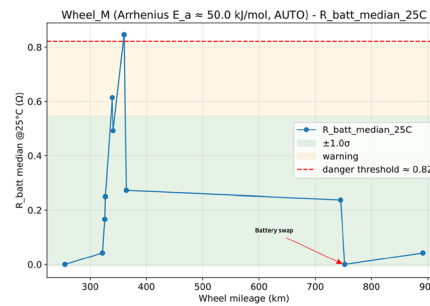


(d) 95th-percentile voltage sag $sag_{95\%}$ with CUSUM-based change detection for Wheel_P_bad.

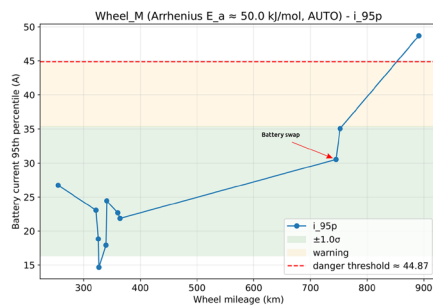
Fig. 11 CUSUM charts for the stylized synthetic Wheel P trajectory (global ageing mode). The early-life logs of real Wheel P are used as reference, and late-life points where the synthetic indicators depart from this baseline are highlighted as intra-wheel regime changes. The goal is to evaluate the responsiveness of the simple CUSUM configuration under controlled knee-like degradation patterns, not to estimate real-world failure times



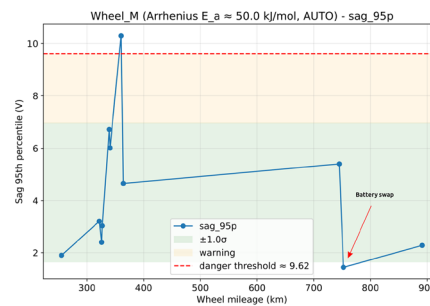
(a) Median equivalent pack resistance ELP_{median} vs wheel mileage for Wheel M (intra-wheel indicator).



(b) Battery-dominated pack resistance $R_{batt,median@25^{\circ}C}$ vs wheel mileage for Wheel M.



(c) Battery current 95th percentile i_{95p} vs wheel mileage for Wheel M.



(d) 95th-percentile voltage sag $sag_{95\%}$ vs wheel mileage for Wheel M.

Fig. 12 Example of a wheel (Wheel M) for which the battery pack was replaced during the observation window, according to user maintenance records. The replacement appears as a step change in equivalent pack resistance, battery-dominated resistance and sag metrics, followed by a new stable regime. This illustrates how EUC_SOH highlights large intra-wheel shifts in pack-level indicators, even though it does not resolve cell-level phenomena

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Author contributions

G.LBL. is the sole author of this manuscript and the only person responsible both for experimenting and writing.

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Data availability

All data and code available at the the following Zenodo repository <https://doi.org/10.5281/zenodo.21861254>

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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